

Language as Data - Assignment 3

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Deliverable 1: SST-2 Task Analysis

Task Overview

Objective: Classify movie review sentences as **positive** or **negative** sentiment.

SST-2 (Stanford Sentiment Treebank - Binary)

- **Source:** GLUE Benchmark
- **Domain:** Movie review sentences from Rotten Tomatoes
- **Task:** Binary sentiment classification
- **Labels:**
 - 0 → Negative
 - 1 → Positive

Dataset Statistics

Dataset Sizes

Split	Number of Examples
Training	67,349
Validation	872
Test	1,821

Class Distribution

Split	Negative (%)	Positive (%)	Balance
Training	44.22%	55.78%	Slight positive bias
Validation	49.08%	50.92%	Nearly balanced

Sentence Length Statistics

Split	Average	Median	Minimum	Maximum	Std. Deviation
Training	9.41	7	1	52	8.07
Validation	19.55	19	2	47	8.76

Observation:

Validation sentences are significantly longer than training sentences

(19.55 vs 9.41 words on average).

Task Examples

Positive Examples (Label = 1)

- “a gorgeous, witty, seductive movie.”
- “a tender, heartfelt family drama.”
- “a fast, funny, highly enjoyable movie.”

Negative Examples (Label = 0)

- “unflinchingly bleak and desperate”
- “it’s slow — very, very slow.”
- “contains no wit, only labored gags”

Selected Instances

10 Easy Instances

Index	Sentiment	Sentence	Reason
1	Negative	“unflinchingly bleak and desperate”	Strong negative words
4	Negative	“it’s slow — very, very slow.”	Emphatic repetition

29	Negative	"i had to look away — this was god awful."	Explicit disgust
32	Positive	"a gorgeous, witty, seductive movie."	Multiple positive adjectives
55	Positive	"a tender, heartfelt family drama."	Warm positive descriptors
67	Positive	"a fast, funny, highly enjoyable movie."	Clear praise
68	Positive	"good old-fashioned slash-and-hack is back!"	Enthusiastic tone
69	Negative	"definitely one to skip, even for fanatics."	Explicit negative advice
6	Negative	"a sometimes tedious film."	Clear negative descriptor
76	Negative	"the movie fails to live up..."	Explicit failure wording

Criteria:

Short sentences, strong sentiment words, no negations, simple structure.

10 Difficult Instances

Index	Sentiment	Sentence	Reason
13	Positive	"root for them... though closer to pity."	Mixed sentiment signals
20	Negative	"admirable look... but uneven tone..."	Contrasting sentiment
45	Negative	"little beyond momentary joys..."	Subtle negative phrasing
66	Positive	"if you're hard up for... this is your ticket."	Backhanded praise
78	Positive	"nothing outstanding... but good enough..."	Mixed quality assessment

89	Positive	“plot’s crazy... but held my interest...”	Contrasting clauses
90	Positive	“scattershot... but brilliant.”	Negative + positive mix
91	Positive	“hardly a masterpiece, but...”	Negation complicates sentiment
93	Positive	“failure... glorious failure.”	Paradoxical phrasing
33	Negative	“if movie succeeds... too self-conscious...”	Complex conditional structure

Criteria:

Mixed sentiment, negations, contrasting clauses, paradoxical phrasing, and complex sentence structures.

Annotation Analysis

Annotator Characteristics

- **Platform:** Amazon Mechanical Turk
- **Expertise:** Non-expert crowdworkers
- **Annotators per instance:** 3–5
- **Demographics:** Primarily native English speakers

Annotation Guidelines and Process

- **Original scale:** 5-point sentiment scale
 - 0 → Very Negative
 - 4 → Very Positive
- **Guidelines:**
 - Intuitive sentiment judgment
 - Example-based anchoring (e.g., “terrible” = very negative)
- **Binary Conversion:**
 - Labels 0–1 → Negative (0)
 - Labels 3–4 → Positive (1)
 - Label 2 (neutral) → Removed
- **Conflict Resolution:** Majority voting / score averaging

Dataset Quality Assessment

SST-2 is a well-established benchmark with significant strengths: large scale (67K+ examples), real-world movie reviews, multiple annotators reducing bias, and balanced classes. However, limitations include lost nuance from binary conversion, domain specificity to movie reviews, crowdworker quality variability, context-removed sentence fragments, and temporal bias (2005-2013). The distributional shift between training (9.41 avg words) and validation sets (19.55 avg words) is notable. Based on manual analysis, the 10 easy instances have clear sentiment indicators and simple structure, while the 10 difficult instances exhibit genuine linguistic complexity through mixed sentiment, negations, and paradoxical phrasing. Some difficult instances may be mislabeled due to genuine ambiguity. Overall, SST-2 serves well for basic sentiment classification benchmarking but has limitations in nuanced understanding and domain transfer.

Deliverable 2: Fine-tuning Analysis Report

Task Overview

The objective of this deliverable was to fine-tune GPT-2 on the **SST-2** (Stanford Sentiment Treebank) task from the GLUE benchmark. We compared the performance of a pre-trained GPT-2 model against a randomly initialized version and explored parameter-efficient fine-tuning strategies, including LoRA and Partial Fine-tuning.

1. Model Performance Comparison

We evaluated four distinct variants of the GPT-2 architecture. The standard pre-trained model achieved the highest overall validation accuracy, highlighting the significant advantage of transfer learning.

Table 1: Overall Performance Metrics

Variant	Validation Accuracy	Validation Loss
Standard (Pre-trained)	91.97%	0.3192
Random Init (No Pre-training)	80.05%	0.5444
LoRA Fine-tuning	90.14%	0.3210
Partial Fine-tuning	89.22%	0.3142

Key Observations:

- Impact of Pre-training:** Utilizing pre-trained weights resulted in an absolute accuracy improvement of **11.93%** over the randomly initialized model.
- Efficiency vs. Performance:** LoRA achieved a competitive accuracy of **90.14%** while only training **0.24%** of the total parameters (296,448 params), demonstrating high efficiency.

- **Partial Fine-tuning:** Freezing the first 8 layers and only training the last 4 resulted in a slight performance dip compared to full fine-tuning but maintained a lower validation loss (0.3142).

2. Performance on Selected 20 Instances

We analyzed the performance of all four variants on 20 manually selected instances (10 "Easy", 10 "Difficult").

Table 2: Accuracy by Instance Difficulty

Variant	Easy Accuracy	Difficult Accuracy	Overall Accuracy
Standard Pre-trained	90.00%	80.00%	85.00%
Random Init	90.00%	80.00%	85.00%
LoRA	100.00%	80.00%	90.00%
Partial Fine-tuning	100.00%	80.00%	90.00%

3.3 Instance-Level Analysis

Unanimous Success (13/20 instances)

All four models correctly classified 13 instances, including:

- Clear negative sentiment: "This is a terrible movie," "Boring and predictable"
- Clear positive sentiment: "Absolutely brilliant!" "Wonderfully entertaining"
- Negation with negative meaning: "Not the worst movie I've seen"

Unanimous Failure (1/20 instances)

"It's so bad it's good." (True label: Positive)

- All models predicted: Negative (0)
- **Reason:** This instance contains ironic/sarcastic language where "bad" is used positively
- **Analysis:** The phrase "so bad it's good" is a cultural idiom indicating enjoyment despite (or because of) poor quality. Models failed to capture this non-literal sentiment

Mixed Results (6/20 instances)

Examples where some models succeeded:

- "I loved every minute of it." (3/4 correct)
- "Horribly acted." (3/4 correct)
- "A guilty pleasure." (3/4 correct)

Pre-training advantage: The pre-trained model correctly classified "Horribly acted" and "Better than average, but still mediocre" while random initialization failed, demonstrating transfer learning benefits.

4. Conclusions

4.1 Key Findings

1. **Pre-training is essential:** Provides 11.93 percentage point improvement over random initialization
2. **Parameter-efficient methods are viable:** LoRA achieves near-standard performance with 0.24% parameters
3. **Models struggle with complex language:** 80% accuracy on difficult instances across all variants
4. **Sarcasm remains challenging:** Unanimous failure on ironic sentiment
5. **LoRA excels on simple cases:** Perfect accuracy on easy instances

Table 3: Training Logs Summary

Variant	Training Time	Final Training Loss
Standard	30:11	0.3255
Random Init	30:21	0.4524
LoRA	20:34	0.3466
Partial	16:28	0.3825

Deliverable 3 - Multilingual Model

Brief Overview

- Model from HuggingFace 'ai-forever/mGPT'
 - GPT-like models with 1.3 billion parameters trained on 61 languages from 25 language families using Wikipedia and Colossal Clean Crawled Corpus.
 - Model is not instruction fine-tuned
 - `zero_shot_prompt = f"English: {original_english}\nFrançais:"`
- Europarl parallel corpus: extracted from the proceedings of the European Parliament. It includes versions in 21 European languages
 - We use English - French

Some Translations

English	Original French	Generated French
Sustainable agriculture and biogas: review of EU legislation (	Agriculture durable et biogaz: nécessité de revoir la législation communautaire (	Agriculture biologique et biogaz: revue de la législation européenne (
rapporteur. - Mr President, we agreed to change paragraph 41 and not to propose a specific biogas EU directive but to integrate it into the draft renewable energy directive.	rapporteur. - (EN) Monsieur le Président, nous avons accepté de modifier le paragraphe 41 et de ne pas proposer de directive communautaire spécifique sur le biogaz, mais d'intégrer ce concept dans le projet de directive sur les sources d'énergie renouvelables.	rapporteur. - Monsieur le Président, nous avons convenu de modifier le paragraphe 41 et non de proposer une directive spécifique biogaz mais de l'intégrer dans le projet de directive renouvelable énergétique.
The ratification and implementation of the updated ILO conventions (vote)	Ratification et mise en œuvre des conventions de l'OIT mises à jour (vote)	La ratification et l'application des conventions de l'Organisation internationale du travail (vote)

Inference on corpus samples & BLEU calculation

- 70 maximum tokens allowed for generations
 - Some translations do not fit into this limit (48 of them are truncated)
- Evaluation over 500 translations
- Calculated length ratio (generated translations / references)
- Applied smoothing to avoid 0 BLEU scores

Overall Results

Average BLEU Score	0.2614
Median BLEU Score	0.2353
Average BLEU Score (excluding truncated translations)	0.2652
Median BLEU Score (excluding truncated translations)	0.2402
zero BLEU % (count)	0.002 (1)
Average Length Ratio	0.92
Median Length Ratio	0.93

5-Best BLEU Scores

English	Original French	Generated French	BLEU Score
- Before the vote on paragraph 41:	Avant le vote sur le paragraphe 41:	- Avant le vote sur le paragraphe 41:	1.0
The debate is closed.	Le débat est clos.	Le débat est clos.	1.0
That is very clear.	C'est très clair.	C'est très clair.	1.0
The debate is closed.	Le débat est clos.	Le débat est clos.	1.0
The answer is simple.	La réponse est simple.	La réponse est simple.	1.0

5-Lowest BLEU Scores

English	Original French	Generated French	BLEU Score
The next item is the joint debate on	L'ordre du jour appelle en discussion commune les rapports suivants:	La prochaine chose est le débat sur	0.0
It means removing, or at least toning down, the criticisms as far as possible in order not to upset anybody.	Cela signifie l'élimination, ou à tout le moins l'adoucissement, des critiques dans toute la mesure du possible, afin de ne choquer personne.	Cesser de critiquer.	0.0004
They were short-changed at the negotiations and did not even notice.	Ceux-ci se sont fait "avoir" lors des négociations, et ne l'ont même pas remarqué.	Ils n'étaient pas changés.	0.0047
The result was unbalanced draft resolutions which we in the European Union were unable to support	Le résultat en a été des projets de résolution déséquilibrés que l'Union européenne n'a pas été en mesure de soutenir.	Le résultat n'était pas uniforme.	0.006
Democratic authorisation and the role of the European Parliament are not in any doubt, and therefore I also insist that the European Parliament and the Member States be given information at every stage, from the intention of a Member State to start negotiations with a third country through to conclusion of an agreement.	La légitimité démocratique et le rôle du Parlement européen ne doivent pas être remis en cause, c'est pourquoi j'ai également insisté pour que le Parlement et les États membres reçoivent des informations à chaque étape, du moment où un État membre décide d'entamer des négociations avec un pays tiers jusqu'à la conclusion d'un accord.	Il n'y a pas de doute sur la légitimité de l'autorisation d'exportation, mais il n'y a pas de doute sur la légitimité de l'autorisation d'importation.	0.0079

Strengths and Weaknesses of BLEU

Strengths

- Fast and easy to compute
- Reproducible and widely adopted
- Useful for comparing models or decoding settings
- Captures n-gram overlap, reflecting some lexical fidelity

Weaknesses

- Sensitive to exact word choice and order
- Penalizes synonyms or paraphrases heavily
- Poorly handles very short sentences
- Ignores semantic correctness and fluency